COMSATS University Islamabad

Department of Mathematics

Course Outline

MTH-262 Statistics and Probability Theory (BDS-IV)

Lectures: Monday and Tuesday: **Instructor:** Abdul Wakeel

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Overview:

Statistics is the study of the collection, organization, analysis, and interpretation of the data. It deals with the planning of data collection in terms of the design of surveys and experiments and decision making based on data.

It is a math that manipulates data that was collected through observation, so that we can better understand what that data means. With statistics, scientists and other analysts have the opportunity to make inferences as to what the data means. Statisticians contribute to scientific inquiry by applying their mathematical and statistical knowledge to the design of surveys and experiments; the collection, processing, and analysis of data; and the interpretation of the results. Statisticians may apply their knowledge of statistical methods to a variety of subject areas, such as biology, economics, engineering, medicine, public health, psychology, marketing, education, and sports. Many economic, social, political, and military decisions cannot be made without statistical techniques

Objectives:

At the end of this course the students of Computer Sciences and Bio Informatics will be able to understand data analysis, modeling, and predictions in their respective fields. The content of this course covers all the descriptive statistics and probability models along with some basic touch of regression analysis.

Course Outline

Introduction to statistics and statistical methods, Frequency Distributions & Representation of data, Measure of Central tendency, Measures of Dispersion, Probability theory, Counting Rules, Conditional Probability, Law of total probability and Bays Rule, Concept of Discrete and Continuous Random variable, Cumulative distributions, Joint probability Distributions, Uniform, Binomial and, Poisson, & Geometric Distributions, Uniform & Normal Distribution, Gamma, Exponential distributions, Simple linear Regression and fitting of Curves. Correlation study. Testing about population Mean, proportion for one sample and two samples. Confidence interval for population Mean, proportion for one sample and two samples.

Books:

Textbook: Ronal E. Walpole & Raymond H. Myres: Probability and Statistics for Engineers and Scientists (8th Ed.)

Reference Books:

- Ronald E. Walpole: Introduction to Statistics (3rd Ed.)
- Neil A. Weiss: Introductory Statistics (10th Ed.)

Grading Policies:

Your final grade will be based on:

Assignments	10%
Quizzes	15%
Mid Exam	25%
Final exam	50%

Please note that the policies stated above are non-negotiable.

Attendance

At least 80% attendance is required to get a chance to pass this course.